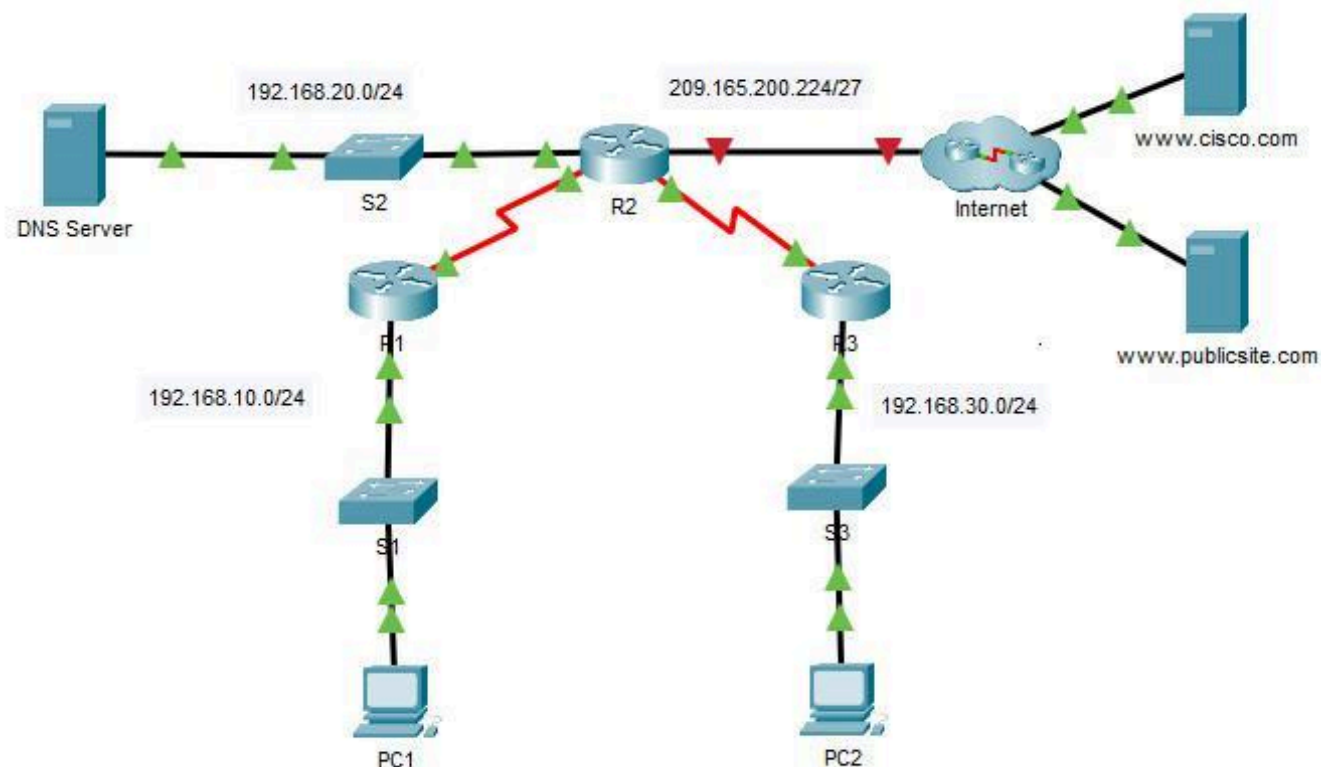


Packet Tracer - Configure DHCPv4

NAME: Carbonel, Nicole D.G.	Date: May 05, 2026
SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

TOPOLOGY:



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.10.1	255.255.255.0	N/A
	S0/0/0	10.1.1.1	255.255.255.252	N/A
R2	G0/0	192.168.20.1	255.255.255.0	N/A
	G0/1	DHCP Assigned	DHCP Assigned	N/A
	S0/0/0	10.1.1.2	255.255.255.252	N/A
	S0/0/1	10.2.2.2	255.255.255.252	N/A
R3	G0/0	192.168.30.1	255.255.255.0	N/A
	S0/0/1	10.2.2.1	255.255.255.0	N/A
PC1	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
PC2	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
DNS Server	NIC	192.168.20.254	255.255.255.0	192.168.20.1

Objectives

Part 1: Configure a Router as a DHCP Server

Part 2: Configure DHCP Relay

Part 3: Configure a Router as a DHCP Client

Part 4: Verify DHCP and Connectivity

Scenario

A dedicated DHCP server is scalable and relatively easy to manage but it can be costly to have one at every location in a network. However, a Cisco router can be configured to provide DHCP services without the need for a dedicated server. As the network technician for your company, you have been assigned the task of configuring a Cisco router as a DHCP server. You are also required to configure the edge router as a DHCP client so that it receives an IP address from the ISP network.

Instructions

Part 1: Configure a Router as a DHCP Server

Step 1: Configure the excluded IPv4 addresses.

Addresses that have been statically assigned to devices in the networks that will use DHCP must be excluded from the DHCP pools. This avoids errors associated with duplicate IP addresses. In this case the IP addresses of the R1 and R3 LAN interfaces must be excluded from DHCP. In addition, nine other addresses are excluded for static assignment to other devices such servers and device management interfaces.

- a. Configure **R2** to exclude the first 10 addresses from the R1 LAN.

```
R2(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

- b. Configure **R2** to exclude the first 10 addresses from R3 LAN.

```
R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
R2(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.10
R2(config)#
```

Step 2: Create a DHCP pool on R2 for the R1 LAN.

- a. Create a DHCP pool named **R1-LAN** (case-sensitive).

```
R2(config)# ip dhcp pool R1-LAN
```

- b. Configure the DHCP pool to include the network address, the default gateway, and the IP address of the DNS server.

```
R2(dhcp-config)# network 192.168.10.0 255.255.255.0
```

```
R2(dhcp-config)# default-router 192.168.10.1
```

```
R2(dhcp-config)# dns-server 192.168.20.254
```

```
R2(config)#
R2(config)#ip dhcp pool R1-LAN
R2(dhcp-config)#network 192.168.10.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.10.1
R2(dhcp-config)#dns-server 192.168.20.254
R2(dhcp-config)#
```

Step 3: Create a DHCP pool on R2 for the R3 LAN.

- Create a DHCP pool named **R3-LAN** (case-sensitive).
- Configure the DHCP pool to include the network address, the default gateway, and the IP address of the DNS server. Refer to the Addressing Table.

```
R2(config)#
R2(config)#ip dhcp pool R3-LAN
R2(dhcp-config)#network 192.168.30.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.30.1
R2(dhcp-config)#dns-server 192.168.20.254
R2(dhcp-config)#exit
R2(config)#
```

Part 2: Configure DHCP Relay

Step 1: Configure R1 and R3 as a DHCP relay agent.

For DHCP clients to obtain an address from a server on a different LAN segment, the interface that the clients are attached to must include a helper address pointing to the DHCP server. In this case, the hosts on the LANs that are attached to R1 and R3 will access the DHCP server that is configured on R2. The IP addresses of the R2 serial interfaces that are attached to R1 and R3 are used as the helper addresses. DHCP traffic from the hosts on the R1 and R3 LANs will be forwarded to these addresses and processed by the DHCP server that is configured on R2.

- Configure the helper address for the LAN interface on R1.

```
R1(config)# interface g0/0
R1(config-if)# ip helper-address 10.1.1.2
```

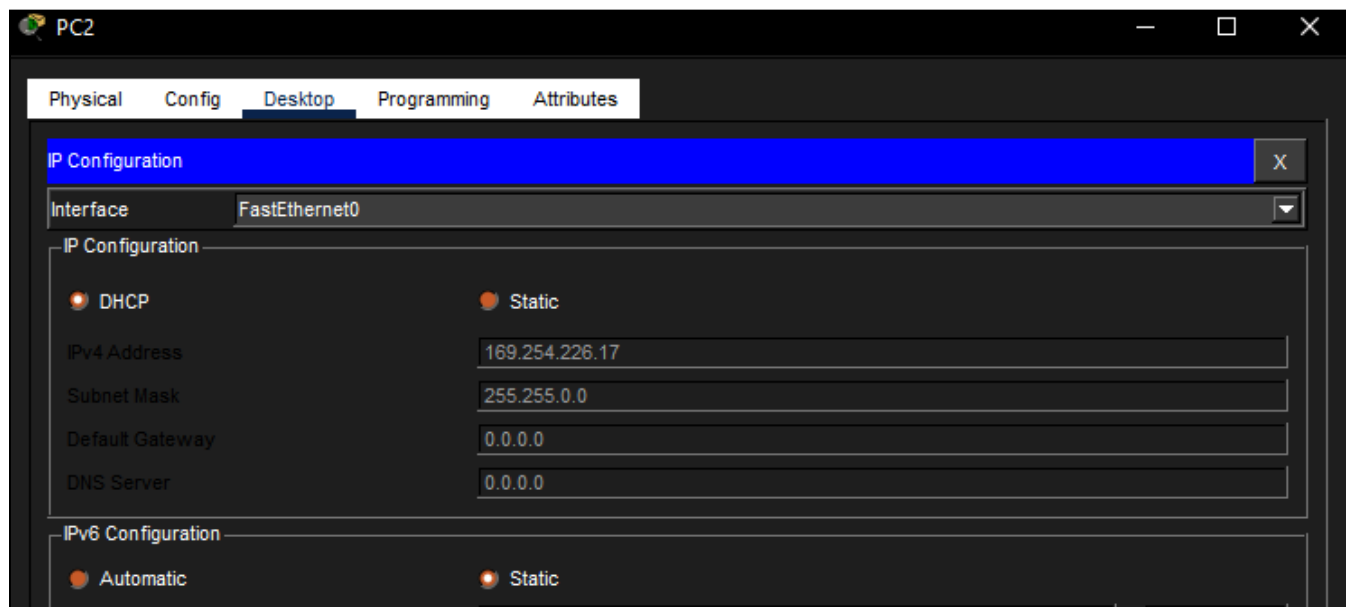
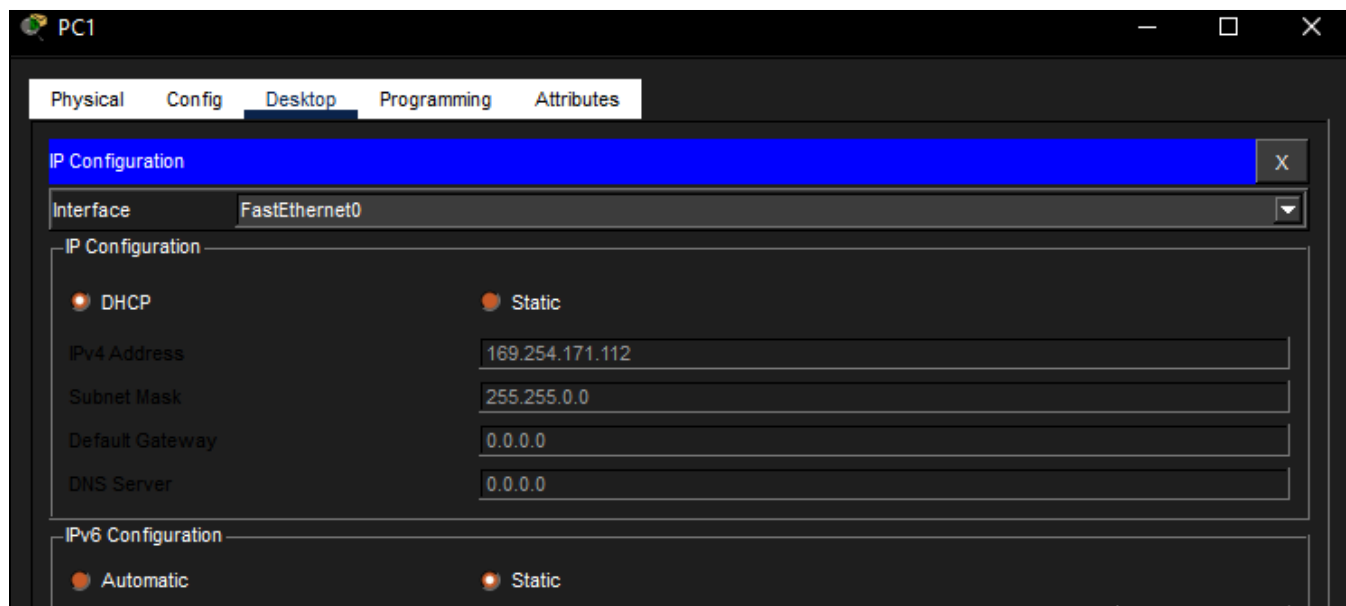
```
R1>
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ip helper-address 10.1.1.2
R1(config-if)#exit
R1(config)#
R1(config)#
```

- Configure the helper address for the LAN interface on R3.

```
R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int g0/0
R3(config-if)#ip helper-address 10.2.2.2
R3(config-if)#exit
R3(config)#
```

Step 2: Configure hosts to receive IP addressing information from DHCP.

- Configure hosts PC1 and PC2 to receive their IP addresses from a DHCP server.
- Verify that the hosts have received their addresses from the correct DHCP pools.



Part 3: Configure a Router as a DHCP Client

Just as a PC is able to receive an IPv4 address from a server, a router interface has the ability to do the same. Router **R2** needs to be configured to receive addressing from the ISP.

- a. Configure the Gigabit Ethernet 0/1 interface on **R2** to receive IP addressing from DHCP and activate the interface.

```
R2(config)# interface g0/1
R2(config-if)# ip address dhcp
R2(config-if)# no shutdown
```

```
R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int g0/1
R2(config-if)#ip address dhcp
R2(config-if)#no shut
R2(config-if)#
```

Note: Use Packet Tracer's **Fast Forward Time** feature to speed up the process.

- b. Use the **show ip interface brief** command to verify that R2 received an IP address from DHCP.

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 192.168.20.1    YES manual up              up
GigabitEthernet0/1 unassigned      YES DHCP    up              up
GigabitEthernet0/2 unassigned      YES unset   administratively down down
Serial0/0/0         10.1.1.2        YES manual up              up
Serial0/0/1         10.2.2.2        YES manual up              up
Vlan1               unassigned      YES unset   administratively down down
R2#
R2#
```

Part 4: Verify DHCP and Connectivity

Step 1: Verify DHCP bindings.

```
R2# show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.10.11	0002.4AA5.1470	--	Automatic
192.168.30.11	0004.9A97.2535	--	Automatic

```
R2>ena
R2#show ip dhcp binding
IP address      Client-ID/  
Hardware address    Lease expiration    Type
192.168.10.11    0090.0C63.AB70      --                  Automatic
192.168.30.11    0002.17CA.E211      --                  Automatic
R2#
```

Step 2: Verify configurations.

Verify that **PC1** and **PC2** can now ping each other and all other devices.

